

Towards Responsible Data Practices in Digital Health: A case study of an open source community's journey

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ABSTRACT

In recent years, privacy and data rights have garnered growing attention in public discourse, policy making, and scholarly research. New data protection laws are being rolled out globally to codify data rights and ensure individual control over how personal data is shared and used. This evolving landscape presents several opportunities and challenges for healthcare. In this case study, we outline a design research agenda that emerged from the practical needs of an open source community focused on digital health software for community health in low- and middle-income countries. We situate this case study in the global landscape of data regulations, and the call for responsible data practices that go above and beyond regulatory compliance. We share findings from the formative stages of our multi-stage design process, which include a scoping literature review and a reframing of institutional policies and procedures. A primary contribution of our case study is that it offers an example of the institutional 'pre-work' necessary to make sense of the complex data protection landscape, and to chart a path forward for designing software that better supports responsible data practices. This case study also articulates the important role for digital health designers and implementers in operationalizing patient data rights.

CCS CONCEPTS

• **Human computer interaction**; • **Health care information systems**; • **Law**;

KEYWORDS

data rights, responsible data, digital health, informed consent, explicit consent, privacy, case study, ethics, community health, GDPR

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1 INTRODUCTION

High-profile scandals involving health [24] and humanitarian data [1], and a wider discourse on surveillance capitalism [28, 33], have thrown into sharp relief the human stakes of what might otherwise seem a relatively dry legal topic: personal data rights. Across the globe, personal data rights are increasingly recognized in practices and policies, including the European Union's General Data Protection Regulation (GDPR) [34]. Indeed, individual data rights have been recognized as a critical component of a rights-based approach to digital citizenship [31, 37]. Many countries in the global south, such as Kenya and Uganda, have adopted new GDPR-influenced legal frameworks to better protect their citizens' data rights.

Today, the impact of these protections is most commonly experienced online, where consumers interact with new website pop-up menus to select how their data can be processed. Consumers are asked to opt-in to data use for non-essential purposes, with a rather crisp line between using data for essential and nonessential website functions. The definition of essential becomes fuzzier when applied to healthcare since health data privacy rights require full and transparent informed consent processes, which go beyond the 'accept these cookies' warnings on websites. For example, in the context of health data, patients must be able to recognize the extensive amount of data that may be captured during their care—including genomic, biometric, and diagnostic data, and a myriad of other measurements, symptoms and diagnoses. They must also understand the risks and benefits of data use, both in their immediate care and in a range of potential research or public health initiatives. This includes the risks of not using patient data and implications for one's

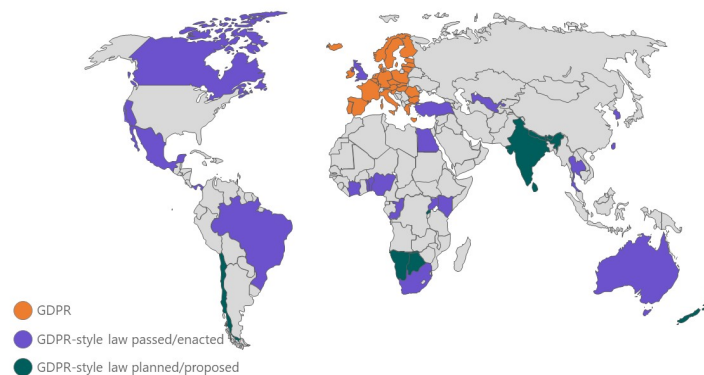


Figure 1: Map Illustrating GDPR and GDPR-style laws enacted or planned globally as of October 2020 [10, 38]

personal as well as population health. The process of consent must be easy for patients to understand, simple for frontline healthcare providers to support, and feasible for digital health stakeholders to implement and maintain. Thus, extending data privacy protections in routine healthcare settings warrants a nuanced approach, careful design, and rigorous research.

This case study describes the formative research stages of a multi-phase, human-centered design and research project to create such data protections. The non-profit organization Medic Mobile set out to develop data policies and tools to support its work with the Community Health Toolkit open-source project. The aim was to better reflect an evolving global policy landscape, while creating data privacy and consent protections that meet the needs of health workers and communities in a range of lower-income settings. We approach this work with an acute focus on practical implementation challenges, as well as on bridging the gap between regulatory documents, lofty ideals, and day-to-day data practices.

This case study starts with an introduction to the complex landscape of responsible data practices, data privacy protections and the implementation challenges faced by health systems in the global south. It then details a scoping literature review and process for revamping institutional policies and procedures to cultivate more responsible data practices. The case study ends with a discussion of how to create and study further digital infrastructures that support rigorous data privacy practices, particularly in low resource settings.

2 CONTEXT

2.1 General Data Protection Regulation Overview

Data privacy laws are rapidly expanding around the world, with 142 countries having laws in place currently, 62 of which were established in the last decade [10]. While the European Union’s Charter on Fundamental Rights, written in 2000, acknowledged the protection of personal data as a fundamental human right, it was in 2016 that GDPR operationalized data privacy rights for European Union citizens and residents [9].

GDPR regulates how data controlling and processing organizations may work with personal data—defined as any information that can be used individually or in a dataset to identify an individual person. GDPR is becoming a global standard; it has had major influence on new or updated data regulations in a growing number of countries, including India, Panama, Kenya, Uganda, Barbados, Thailand, Brazil, Nigeria, and South Africa (Figure 1).

Consent, in the context of healthcare, is a highly regulated yet nuanced process that balances individual rights with the needs of scientific research and population health. The concept of consent outlined in GDPR requires that when “the data subject’s consent is to be given following a request by electronic means, the request must be clear, concise and not unnecessarily disruptive to the use of the service for which it is provided” [35]. If data is being shared and used for a variety of reasons, per GDPR, then “consent should be given for all of them” [35]. For health-related data, GDPR points to an even higher level of consent called “explicit consent”. Although the parameters of “explicit” are not clearly articulated in GDPR, it is interpreted to mean that consent for processing healthcare data requires especially clear and specific language [30]. GDPR has also brought global attention to the notion of a “right to be forgotten”—the expectation that an individual may request that their personal data be fully deleted from a database. This right requires responsive technology infrastructure where an individual’s preferences must be documented and retrievable, and able to be regularly referenced and deleted at will.

2.2 Responsible Data and the Community Health Toolkit

In the last decade, a growing body of work has established that digital health interventions can improve the coverage, quality, speed, and equity of care in hard-to-reach communities around the world [15, 17]. The Community Health Toolkit (CHT) is an open-source project that designs and builds some of the most widely used software for digital community health in lower- and middle-income settings. Digital health apps built using the CHT have been deployed in over 20 countries in the global south and are now used

by about 30,000 health workers to support up to 1.5 million care activities every month. In addition to enabling care coordination, records management, and task management for health workers, these apps generate datasets that support healthcare delivery, enable advocacy, and facilitate social change. Medic Mobile serves as the technical steward of the CHT open-source community and, in this capacity, supports a range of collaborative, technical and research activities in-line with its organizational values of humanity, creativity, initiative, solidarity, and openness [39].

Being responsible stewards of people's data is critically important to Medic Mobile's mission and to the wider CHT community. At the outset of this project, we adopted the concept of Responsible Data to outline our duty to prioritize and respond to the ethical, legal, social, and privacy-related challenges that come from generating and using data in new ways [37]. The concept of responsible data moves beyond legal compliance and implies that organizations have an obligation to account for unanticipated and unintended consequences of working with data, where we incorporate Parvin and Pollock's recent exploration of the phrase "unintended consequences" [23]. Responsible data practices include prioritizing people's rights to consent, privacy, security, and ownership, as well as implementing values and practices of transparency and openness in the creation of digital tools, continued use, and collection of data. In the global health industry, organizations like Medic Mobile, which work in multiple countries and sites, must now comply with disparate national data regulations, while maintaining a higher institutional commitment to Responsible Data, even in settings with less stringent data protections. As this case study demonstrates, the peer reviewed literature and the practitioner community lack clear examples of how global health organizations might navigate this challenge effectively.

3 APPROACH

To understand how digital tools and technology can best support responsible data practices, we conducted a landscape scoping review and a revamp of Medic Mobile's data protection policies. These are the first two stages of a multi-phase, design research process to uncover best data privacy practices in routine care settings and then to instantiate these practices within open-source digital health software.

3.1 Landscape Scoping Literature Review

We began with a ten-week scoping literature review to get a sense of the current data privacy landscape across multiple disciplines. In contrast with systematic literature reviews, scoping review methods are suited to ambiguous topics for which key search terms may not be apparent at the outset of the study, and they are often time-bound for pragmatic reasons [4, 11]. We explored both the ethical and practical dimensions of collecting, sharing, and using routine health data with the goal of identifying best practices. We identified two major areas of interest: i) informed consent and ii) data sharing and data use including secondary analyses and research, and searched for these themes in both empirical research and policy papers.

Given Medic Mobile's global footprint, we aimed to uncover interventions from low- and middle-income settings while gaining a better understanding of current European Union and United States

Table 1: Scoping Literature Review Paper Summary

Database	Total Papers Identified ¹	Total Papers Scanned ¹	Final Papers Reviewed
Hein Online	187	101	58
PubMed	724	63	32
Cited Literature	n/a	n/a	12
ACM	21	21	8
Global Health	23	13	4
Annual Reviews	18	5	5
Reviewer Recommended	5	5	5

policies given their influence in global health. We timebound our scoping review (April-June 2020) based on University of Washington's academic calendar. Three PhD students from the Department of Human Centered Design and Engineering led the review process. Noting that data ethics and informed consent spans several academic disciplines, we met with five University of Washington librarians with specific specialties in public policy, law, health, information studies, and international studies to better understand where and how to search existing literature. The librarians identified 22 different relevant databases and, of these, we pulled literature from the five most relevant databases for empirical studies and policy papers: ACM, PubMed, Global Health, HEIN Online, and Annual Reviews. Most of the papers came from HEIN online and PubMed. In PubMed our search was limited by MeSH terms, which excludes more recent publications that have not been formally indexed, and we sorted by "best match" and reviewed the top 60 articles (Appendix A.1). We also scanned the literature cited in the papers reviewed and included 12 additional, relevant papers. During this manuscript's review process, one reviewer recommended five papers to review, which led us to several additional sources included in the table below.

3.1.1 Landscape Scoping Findings:

Informed Consent. Both the policy and empirical literature demonstrate that healthcare's current conception of consent is strongly rooted in consent processes for research studies, where consent is for a specific study collecting data for a defined scope, say, to test a drug or an intervention. Empirical studies of research consent focus heavily on informed choice: ensuring potential participants understand the specific risks and benefits of participation, while ensuring a non-coercive consent process that does not coerce.

One major theme was recognizing the differences between broad consent versus more specific opt-in consent procedures for sharing data captured during a study [5, 13, 29]. As Bull and colleagues note in their multisite qualitative study of research data practices across five countries (India, Thailand, Vietnam, South Africa, and Kenya): "Many arguments in favor of broad consent were pragmatic, with researchers referring to the difficulty and expense of recontacting

¹This number likely includes duplicates as we conducted different searches in each database and did not deduplicate across searches. Search terms are included in Appendix 1.

research participants” [5]. Two papers also note that recontacting a patient after a research study to request secondary data use could conceivably cause more harm due to accidental, personal health information disclosure [5, 18].

Staff training and seniority are also recognized as a major barrier to consent in several studies [8, 14, 25, 26]. Denny and colleagues observe differences based on the seniority of the personnel conducting informed consent in South Africa [8]. Similarly, Sanderson and colleagues recommend deeper training for healthcare professionals involved in a genomic study, where training should emphasize listening, open-ended questions, and giving participants time to think to ensure supportive decision making [26]. Jao and team note that researchers and frontline staff were concerned with potential for consent to become “a ‘tick box’ exercise” due to challenges in explaining data sharing [14]. Ramos and colleagues recommend hiring a specialized cadre of personnel to support consent for data sharing at a New York City HIV care clinic, given a need for expertise in the intricacies of data sharing and needing a dedicated patient liaison to the process [25]. Notably, the majority of empirical studies focus on informed consent during research-related activities and none responded to GDPR-style data privacy guidelines.

Data Sharing and Use. Secondary uses of research data comprise an area of vibrant discussion in the literature, with particular emphasis on consent in the new era of big data and use of personal data to create aggregate or de-identified datasets [3, 7, 22]. Several studies also outline the need for data governance models [5, 19, 27]. Bull and colleagues found variety in data governance models across the five countries, where “views differed between sites about the appropriate composition of governance groups. In some settings, it was considered appropriate for scientific and research ethics committees to review data-sharing proposals, while in others, the value of establishing novel data-sharing committees or appointing data managers was considered. Views also differed about which stakeholder interests should be represented in governance processes and how best to do so” [5]. Murtagh and colleagues offer a participatory model of data governance to support managing longitudinal study data in the UK where they advocate for study participant representation in data access committees [19]. There were many papers discussing the importance of data sharing for the public or greater good while noting the tension with the “competitive value of research data, in terms of advancing [research] careers,” [5] an idea echoed for the CHI community in Pal’s 2017 “CHI4Good or Good4CHI” [21].

Scoping Review Evidence Gaps. Overall, we note many gaps in the literature reviewed. Most studies center on exploring elements of informed consent and data sharing in research settings. There was a lack of research about consent in routine care settings, potentially influenced by the newness of GDPR-style legislation. An October 2020 PubMed general search of “explicit consent” and “GDPR” yielded just two relevant results, both published in the last two months [6, 16]. Additionally, many of the empirical studies took place in higher-income countries with larger human resource potential and overall resources; there is an acute dearth of studies that focus on resource constraints and pragmatic opportunities to improve data privacy practices during routine care in low- and middle-income countries.

Our review also found a lack of studies on digitally enabled consent processes. Most of the relevant empirical work alluded to paper-based consent forms, with only one noting that written consent information was transcribed into an electronic registration system [25]. In addition, there was little mention of the technical infrastructures required to operationalize data privacy practices. To be sure, the recent 2020 CSCW Ethics In Design workshop highlighted several new case studies and tools with a goal to “propose and consider new interventions and approaches to ethics in design that go beyond formal checklist- and compliance-oriented approaches” [36]. We also point readers to a new IEEE scoping review on security and privacy practices in digital health applications that calls for a “repository of evaluation methods and design guidelines that would support compliance [and] provide a fine-grained view on incorporating security and privacy for digital health into design process” [20]. Nevertheless, there remains very little empirical work on digitally supported consent practices in the peer-reviewed literature.

Finally, we found it particularly difficult to synthesize literature findings into robust categories, which we hypothesize is emblematic of this nascent area of research. It is possible that an extended review including additional databases would have uncovered further relevant research. Nevertheless, based on our rapid scoping review, it appears there is limited research concerning the design of software that embodies responsible data practices, supports the specific needs of healthcare delivery, and accommodates emerging national privacy policies. The absence of robust literature in this area reinforces the need for our work to build an informed consent process using a human-centered design process, with careful attention to the multi-stakeholder landscape.

3.2 Reframing internal policies:

To begin operationalizing lessons from the scoping literature review, we began reviewing and revising Medic Mobile’s data protection practices and policies. Through dozens of digital health projects over the last decade, Medic Mobile accumulated a patchwork of institutional policies and agreements that reflected the needs of many specific collaborators and projects. As a whole, these policies did not establish a clear, consistent, high bar for a responsible data practice that could be maintained across regulatory environments. Similarly, it had become increasingly tedious to understand and manage compliance with this variety of project agreements. To address this situation, we began to develop infrastructure for internal compliance, with a focus on: i) establishing more uniform expectations for responsible data practice; ii) more thoroughly embracing GDPR-style concepts and language; and iii) designing policies that would be agreeable to Ministries of Health and non-governmental collaborators.

We started this work by thoroughly reading data protection policies across key implementing countries, such as Kenya and Uganda, noting that many of Medic’s collaborations occur in countries that have adopted GDPR-style legislation. The next phase of work included a review of the partner agreements that Medic Mobile and their collaborators sign when they initiate new digital health projects. Like many nonprofits, the organization relies on a

Table 2: Overview of Standard Partner Agreements

Agreement Type	Purpose
Memorandum of Understanding (MOU)	Outline general terms of a partnership [32]
Scope of Work (SoW)	Articulate specific project goals, timelines, and deliverables
Data Sharing and Use Agreement (DUA)	Outline terms of project data sharing and use

set of documents to outline specific project terms and agreements with new partners (Table 2).

We worked to weave in GDPR-style requirements across these agreements and focused on deep revisions to the DUA template. Previously, Medic Mobile's DUA was a standard template where project-specific terms of data sharing and use were outlined in the SoW. We revised the DUA by pulling in language from both the MoU and SoW with the aim of creating a stand-alone DUA that would become the "go-to" source for project data sharing and use. The goal was to make it easier for partners and Medic Mobile staff to understand project data specifics, while also making it simpler to update when projects evolve and data needs change. We also began creating template documents that could be adapted by a broader open-source community. The revamped DUAs outline what project data will be required, how parties will comply with local data protection laws, how data can be shared and accessed, how signatories may use data generated by the project, and how long data will be retained after project completion.

In addition to supporting GDPR compliance at Medic Mobile, we looked across requirements in multiple data protection policies to draft a Data Protection Officer (DPO) job description. The DPO will proactively support data privacy and compliance, as well as facilitate internal capacity building on responsible data policies and practices. The process of writing the job description also resulted in an internal list of organizational responsibilities and recommendations that will further Medic Mobile's work to instill a culture of data responsibility. For example, Medic Mobile will need to outline internal mechanisms for creating a "culture of independence" necessary for conducting GDPR-required Data Protection Impact Assessments of projects. Additionally, the organization will need to reinforce internal tracking systems to capture specific terms in the revamped DUAs. Finally, the DPO will ensure that Medic Mobile's policies and procedures remain in compliance with updated legislation, and that staff at Medic Mobile are trained on best practices for data security, management, and responsible data use.

After working through the different phases of agreements and new human resource requirements, we then turned back to Medic Mobile's overarching Data Privacy and Protection Policy. Here we began to weave in new sections and include additional information on responsible data use, including data privacy and security, in Medic Mobile's onboarding and staff training. In the context of Medic Mobile's work as the technical steward of the CHT, the organization also plans to expand capacity building offerings for technical partners to include guidance on best practices for data

hosting, including considerations around privacy, security, and data sharing. The goal of this work is to outline and model a series of best practices around responsible data that translate to the broader CHT community.

Findings from reframing internal policies and practices. Overall, we found that revamping data privacy policies and responsible data practices is a labor intensive and complex process. The infrastructures of policies, agreements, and human resource requirements are highly interdependent. Many governments have not or have only recently passed GDPR-style legislation and, thus, have not fully operationalized processes and compliance requirements. Navigating this dynamic took deep investment from Medic Mobile, including hiring an external consultant and routine meetings with senior leadership to align on policies and processes. The hope is that the insights and resources that result from this process will make the transition to responsible data practices simpler and less costly for other CHT implementers, as well as the wider digital health community. All documents that we revised are still under legal review, and Medic Mobile plans to share templates generated during this formative phase on the Community Health Toolkit documentation site [40].

4 FRAMING FUTURE DESIGN WORK

This case study showcases the complexity of extending data privacy rights, and yet is just the beginning of the work required to encode these protections. We see a clear need for continued efforts in three overlapping areas: i) new strategies for informed consent in routine care settings; ii) strengthening digital infrastructure to support best practices; and, iii) collaborative design research to test these strategies and disseminate findings. Community health workers (CHWs) play critical roles in health care delivery and patient advocacy in many health systems; we believe that studying how CHWs may support data rights and informed consent could yield insights of relevance to the Community Health Toolkit and other digital health systems around the world.

In the case of informed consent, we will next co-design informed consent modules by engaging with key stakeholders, including data ethicists, government officials, Institutional Review Board members, health implementers, legal counsel, and CHWs. We will then iteratively design informed consent infrastructures through participatory design [12] with the same broad list of stakeholders, adding in more frontline healthcare providers and patient rights organizations. After evaluating their effectiveness, the code and documentation that support these new experiences will be released as part of the Community Health Toolkit open-source project. We aim to share what worked, what failed, and provide guidance on digital infrastructures to help open-source communities operationalize consent in routine care settings using responsible data practices.

This work is part of a small but growing literature at the intersection of implementation science, human-computer interaction, and global health policy and practice. For routine health systems in low- and middle- income settings, addressing the ethical, legal, social, and privacy-related challenges that come from generating and using data in new ways is remarkably complex. This is particularly challenging for institutions working across many jurisdictions that not only seek to comply with the evolving regulatory landscape, but

also implement the strongest data protection standards. Aspiring to higher standards entails working proactively to anticipate negative consequences, create plans to mitigate them, and accepting responsibility when they occur. In our case study, we explored how to expand support for personal data rights, without threatening the fundamental right to dignified and accessible healthcare, the operational practicality of routine care, or the institutional partnerships that underpin global health. A key contribution of our case study is that it offers a transparent example of the institutional 'pre-work' necessary to make sense of the data protection landscape and chart a path forward. Often, HCI and ICT for Development researchers make critical choices about the scope and aims of a design project before the formal application of human-centered design methods [2, 12]. In this case, a literature review and a comprehensive re-framing of institutional policies played essential roles kickstarting a reimagining of how to create digital infrastructures for data rights and responsible data practices.

Of the many dimensions of responsible data practice, consent is the critical link where privacy rights are established and negotiated between people and health systems. This link will rely on new digital infrastructures that can distill the complexity of healthcare data, while, simultaneously, empowering patients to decide how they want their data to be used. The issue of informed consent for routine care settings would benefit from more attention from the HCI community, as there is an unspoken requirement for digital infrastructures that will be highly complex and require rapid evolution to keep pace with data privacy standards. These efforts would benefit greatly from further human-centered research, particularly in communities with lower levels of data literacy, digital literacy, or literacy in general.

The urgency and influence of further work on this topic cannot be overstated. The complex issues of data protection, privacy, and responsible data practices explored in this case study will increase in importance in the coming years, as digitally supported healthcare expands and the resulting data continue to be a vital resource for data science, global health, and design research worldwide. These data are highly personal and specific to communities impacted by legacies of colonization, structural adjustment, and systemic underinvestment, yet data themselves are often treated as a commodity for further experimentation, research, and decision-making in settings in the global north. This type of extractive relationship is mediated through and facilitated by digital tools; yet there is also potential for digital infrastructure to be designed in a manner that enables the data rights of patients in these settings. We believe that computing and design researchers have a critical role to play in furthering responsible data practices, and hope that our case study will nurture more conversations within the CHI community.

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A APPENDICES

A.1 Literature Scoping Review Search Terms

Database	Search 1	Search 2	Total Papers Identified*	Abstracts scanned*	Full Papers Reviewed
Hein Online	((health) AND (data)) AND text:"(data) (sharing)" ~10)	("(health information) (privacy)" ~5 AND text:"(privacy) (sharing)" ~5)	187	101	58
PubMed	("Patients" [MeSH] OR "Health Personnel" [MeSH] OR "Organizations" [MeSH] OR "Ownership" [MeSH]) AND ("Patient Rights" [MeSH] OR "Ethics" [MeSH] OR "Privacy" [MeSH]) AND ("Data Collection" [MeSH] OR "Information Management" [MeSH] OR "Information Storage and Retrieval" [MeSH] OR "Information System" [MeSH] OR "Records" [MeSH] OR "Information Systems" [MeSH] OR "Telemedicine" [MeSH] OR "Algorithms" [MeSH]) AND ("Clinical Study" [PT] OR "Comparative Study" [PT] OR "Evaluation Study" [PT] OR "Multicenter Study" [PT] OR "Resource Guide" [PT] OR "Technical Report" [PT])	("Clinical Study" [PT] OR "Comparative Study" [PT] OR "Evaluation Study" [PT] OR "Multicenter Study" [PT] OR "Resource Guide" [PT] OR "Technical Report" [PT]) AND "data trust"	724	63	32
ACM	"query": {Abstract:(privacy) AND Abstract:(health) AND Abstract:(intervention)} "filter": {Media Format: PDF},{Article Type: Research Article},{ACM Content: DL}		21	21	8
Annual Re-views	ab:(ethics) AND ab:(information sharing) AND ab:(health)		23	13	4